

United International University

Name
(Optional)

ID No.

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Section

.....
Invigilator's
Signature with date

Course Code

Trimester / Semester : Spring / Summer / Fall, 20.....

Name of Exam : Class Test / Mid-term / Final

Date:

3. (a) physical address = 3D1B2 [DS:BX]

$$i) \text{ } \$ \text{ DS } SG = (3D1B2 - 7B85) / 10$$

$$= \frac{3562D}{10}$$

Not valid as SG cannot be fraction.

$$ii) SG = 38BE$$

iii) Not valid.

The highest of SG 7D92 is

$$7D920 + FFFF = 8D91F$$

3D1B2 is out of range.

$$(b) \quad 2^{n_A} \times 8 = 128 \times 2^{20} \times 8$$

$$\Rightarrow 2^{n_A} = 2^7 \times 2^{20}$$

$$\Rightarrow n_A = 27$$

$$2^{n_B} \times 16 = 128 \times 2^{20} \times 8$$

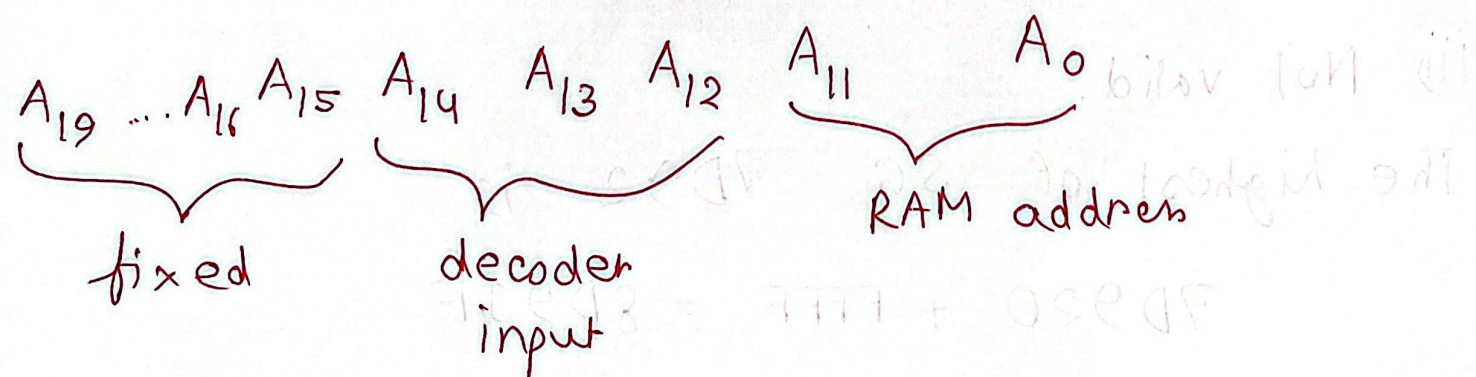
$$\Rightarrow 2^{n_B} = 2^7 \times 2^{19}$$

$$\Rightarrow n_B = 26$$

4. each RAM has $2^{12} = 4 \text{ KB}$ memory

7 RAMs needed.

we use 3:8 decoder



Address map

00000

RAM 0

00FFF

01000

RAM 1

01FFF

02000

RAM 2

02FFF

03000

RAM 3

03FFF

04000

RAM 4

04FFF

05000

RAM 5

05FFF

06000

RAM 6

06FFF



Address map

00000
00FFF

01000
01FFF

02000
02FFF

03000
03FFF

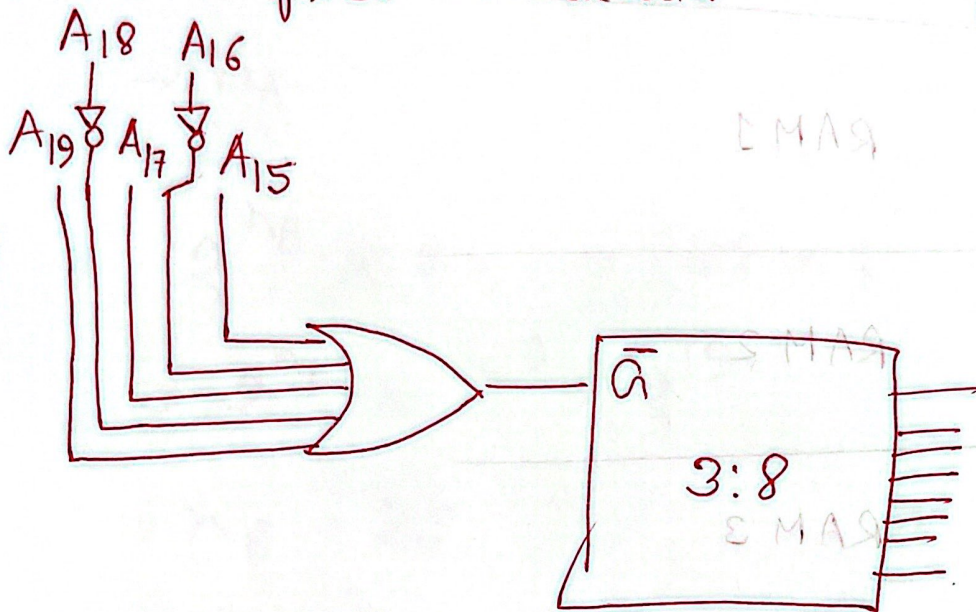
04000
04FFF

05000
05FFF

06000
06FFF

(b) 0101 0000 0000 0000 0000
 0101 0110 1111 1111 1111

fixed decoder RAM address



Address map

50000

RAM 0

50FFF

51000

RAM 1

51FFF

52000

RAM 2

52FFF

53000

RAM 3

53FFF

54000

RAM 4

54FFF

55000

RAM 5

55FFF

56000

RAM 6

56FFF



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5. (a) 5 bits can be used to denote interrupt
type. 4 memory addresses for each type.

So $32 \times 4 = 128$ addresses.

Starting 000

ending 07F